

Fresnel vs Fraunhofer

Slit diffraction: $N = a^2/(\lambda L)$; Cornu spiral; Fresnel integral vs sinc².

FRESNEL VS FRAUNHOFER (1D SLIT)

Far field (small N)

Near field (large N)

Transition

Wavelength λ 550 nm

Half-width a (slit $\pm a$) **0.15** mm

Distance L 1.2 m

****Fresnel number**** $N = a^2/(\lambda L)$ (slit rule-of-thumb): $N \ll 1$ — far field, pattern tends to ****Fraunhofer**** (sinc^2 here); $N \gtrsim 1$ — near field, keep the quadratic phase (****Fresnel**** integral). The ****Cornu spiral**** plots $C(u) + iS(u)$; slit amplitudes are phasor steps between two points on the spiral (qualitative).

SHORTCUTS

- Small $N = a^2/(\lambda L)$: far field; large N : near-field curvature matters
- Compare solid curve to dashed sinc^2

MEASURED VALUES

$$N = a^2/(\lambda L) \quad 0.0341$$

Scale $y \sim \lambda L/a$ **4.40** mm

	Regime (rule of thumb)	Fraunhofer-like (small N)
--	------------------------	------------------------------

About this model

For a 1D slit, the Fresnel–Kirchhoff integral includes the quadratic phase $\exp(i \pi (y-y')^2/(\lambda L))$ across the aperture. The dimensionless Fresnel number $N \approx a^2/(\lambda L)$ (with a the half-width) separates regimes: small N corresponds to far-field propagation where the pattern approaches the Fraunhofer (Fourier) limit, here $\sin^2(\pi W y/(\lambda L))/(\pi W y/(\lambda L))^2$ for width $W = 2a$. Large N means the near-field Fresnel curvature matters. The Cornu spiral traces $C(u)$ and $S(u)$ from the Fresnel integrals; knife-edge and slit amplitudes are phasor steps on this spiral.

Who it's for: After slit diffraction; pairs with Airy disk for circular apertures.

KEY TERMS

- Fresnel number
- Fresnel integral
- Cornu spiral
- Fraunhofer diffraction

How it works

Near field (Fresnel): the quadratic phase $\exp(i \pi (y-y')^2/(\lambda L))$ across the aperture cannot be dropped — the observed pattern is a **Fresnel diffraction** integral. **Far field (Fraunhofer):** for large L (small $N = a^2/(\lambda L)$), the same integral reduces to the **Fourier** (here **sinc**) pattern of the aperture. The **Cornu spiral** plots **C(u)** and **S(u)** from the **Fresnel integrals**; knife-edge and slit amplitudes are **vector** steps on this spiral. This page uses a **direct numerical slit integral** (solid) and overlays the **Fraunhofer sinc²** shape (dashed) for comparison.

Key equations

$$N \approx a^2/(\lambda L) \cdot \text{small } N \rightarrow \text{Fraunhofer}; \text{large } N \rightarrow \text{Fresnel}$$

$$C(u), S(u) = \int_0^u \cos/\sin(\pi t^2/2) dt \cdot \text{Cornu spiral}$$

Frequently asked questions

Why does the dashed curve not match the solid curve?

The dashed line is the ideal Fraunhofer sinc² for the same slit; the solid line is the full Fresnel slit integral. They match best when N is small (L large).

Is N exactly $a^2/(\lambda L)$?

Definitions vary by geometry (slit vs disk; width vs radius). This page uses $N = a^2/(\lambda L)$ with a the slit half-width as a simple scale for teaching.

Related simulators

Continue with similar topics in this category — or [all 50 in Optics & Light](#).

[View category →](#)

SCHOOL

Diffraction

▶ Launch Simulator

Single and double slit with interference patterns.

NEW

SCHOOL

Airy Disk & Rayleigh Limit

▶ Launch Simulator

Circular aperture Fraunhofer pattern; first dark ring; two-point resolution.

NEW

UNIVERSITY / RESEARCH

Bragg's Law (X-ray)

▶ Launch Simulator

$2d \sin \theta = n\lambda$ vs θ ; crystal planes and constructive reflection (schematic).

NEW

UNIVERSITY / RESEARCH

Holography (Recording principle)

▶ Launch Simulator

Toy 2D interference $|E_{\text{ref}} + E_{\text{obj}}|^2$: fringes encode phase; readout diffraction not simulated.

NEW

SCHOOL

Brewster Angle

▶ Launch Simulator

$\tan \theta_B = n_2/n_1$; $R_p \rightarrow 0$; $\theta_i + \theta_t = 90^\circ$; Fresnel R_s, R_p vs θ_i .

NEW

SCHOOL

CD / Grating Rainbow

▶ Launch Simulator

Lines/mm and incidence: scalar grating orders painted as λ -colored fans.